

Problem 2

Problem 1 Determine the following indefinite integrals.

(a)

$$\int (\sec^2(x) + 5) dx$$

Explanation.

$$\int (\sec^2(x) + 5) dx = \tan(x) + 5x + C$$

(b)

$$\int (\sec^2(x) - \sec(x) \tan(x)) dx$$

Explanation. We apply the Sum Rule.

$$\int (\sec^2(x) - \sec(x) \tan(x)) dx = \int \sec^2(x) dx - \int \sec(x) \tan(x) dx = \tan(x) - \sec(x) + C.$$

(c)

$$\int \frac{x + 3x^5}{x^3} dx$$

Explanation. First, we express the function in a more convenient way.

$$\int \frac{x + 3x^5}{x^3} dx = \int \left(\frac{x}{x^3} + \frac{3x^5}{x^3} \right) dx = \int (x^{-2} + 3x^2) dx.$$

Now we apply the Sum Rule and the Constant Multiple Rule.

$$\int \frac{x + 3x^5}{x^3} dx = \int (x^{-2} + 3x^2) dx = \int x^{-2} dx + 3 \int x^2 dx.$$

Now we determine each integral.

$$\int \frac{x + 3x^5}{x^3} dx = \int x^{-2} dx + 3 \int x^2 dx = -\frac{1}{x} + x^3 + C.$$

(d)

$$\int \frac{1 + 2x}{1 + x^2} dx$$

(HINT: Think about the derivative of $\ln(f(x))$).

Explanation. First, we express the function in a more convenient way.

$$\int \frac{1 + 2x}{1 + x^2} dx = \int \left(\frac{1}{1 + x^2} + \frac{2x}{1 + x^2} \right) dx.$$

Now we apply the Sum Rule.

$$\int \frac{1 + 2x}{1 + x^2} dx = \int \frac{1}{1 + x^2} dx + \int \frac{2x}{1 + x^2} dx.$$

Now we determine each integral.

$$\int \frac{1 + 2x}{1 + x^2} dx = \int \frac{1}{1 + x^2} dx + \int \frac{2x}{1 + x^2} dx = \arctan(x) + \ln(1 + x^2) + C.$$

(e)

$$\int \frac{2 + x^2}{1 + x^2} dx$$

Explanation. First, we express the function in a more convenient way.

$$\int \frac{2 + x^2}{1 + x^2} dx = \int \frac{1 + 1 + x^2}{1 + x^2} dx = \int \left(\frac{1}{1 + x^2} + \frac{1 + x^2}{1 + x^2} \right) dx = \int \left(\frac{1}{1 + x^2} + 1 \right) dx.$$

Now we apply the Sum Rule and the Constant Multiple Rule.

$$\int \frac{2 + x^2}{1 + x^2} dx = \int \frac{1}{1 + x^2} dx + \int dx = \arctan(x) + x + C.$$